POINT DESCRIPTION OF THE INVARIANT-RING SPECTRUM M = U/R

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: this chapter's pool has 47 of 128 labels already delivered by the S7 line (S7-gp-* registrations, statement_anchor-prose scan; withdrawn cards excluded); none of the 11 accepted labels of this batch is among them (gp-occupied.json, 2026-10-14).

Source reference. Original equation 03BF: sources/stacks-chow/tags.txt, source lines 4346–4346. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03BJ: sources/stacks-chow/tags.txt, source lines 4350–4350. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00GQ: sources/stacks-chow/tags.txt, source lines 616–616. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01S1: sources/stacks-chow/tags.txt, source lines 2322–2322. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00DS: sources/stacks-chow/tags.txt, source lines 508–508. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03BH: sources/stacks-chow/tags.txt, source lines 4348–4348. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03BK: sources/stacks-chow/tags.txt, source lines 4351–4351. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume $U = \mathrm{Spec}(A)$ and $R = \mathrm{Spec}(B)$ are affine and $s, t : R \to U$ finite locally free. Let $C \subset A$ be as in (03BF). Then $U \to M = \mathrm{Spec}(C)$ has the following properties:*

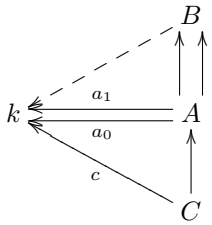
Date: stacks-project:groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *the map on points* $|U| \to |M|$ *is surjective and* $u_0, u_1 \in |U|$ *map to the same point if and only if there exists a* $r \in |R|$ *with* $t(r) = u_0$ *and* $s(r) = u_1$*, in a formula*
$$|M| = |U|/|R|$$
(2) *for any algebraically closed field* k *we have*
$$M(k) = U(k)/R(k)$$

Proof. Since $C \to A$ is integral (Lemma 03BJ) and injective we see that $\mathrm{Spec}(A) \to \mathrm{Spec}(C)$ is surjective, see Algebra, Lemma 00GQ. Thus $|U| \to |M|$ is surjective.

Let k be an algebraically closed field and let $C \to k$ be a ring map. Since surjective morphisms are preserved under base change (Morphisms, Lemma 01S1) we see that $A \otimes_C k$ is not zero. Now $k \subset A \otimes_C k$ is a nonzero integral extension. Hence any residue field of $A \otimes_C k$ is an algebraic extension of k, hence equal to k. Thus we see that $U(k) \to M(k)$ is surjective.

Let $a_0, a_1 : A \to k$ be two ring maps. If there exists a ring map $b : B \to k$ such that $a_0 = b \circ t^\sharp$ and $a_1 = b \circ s^\sharp$ then we see that $a_0|_C = a_1|_C$ by definition. Thus the map $U(k) \to M(k)$ equalizes the two maps $R(k) \to U(k)$. Conversely, suppose that $a_0|_C = a_1|_C$. Let us name this algebra map $c : C \to k$. Consider the diagram



If we can construct a dotted arrow making the diagram commute, then the proof of part (2) of the lemma is complete. Since $s : A \to B$ is finite there exist finitely many ring maps $b_1, \ldots, b_n : B \to k$ such that $b_i \circ s^\sharp = a_1$. If the dotted arrow does not exist, then we see that none of the $a_i' = b_i \circ t^\sharp$, $i = 1, \ldots, n$ is equal to a_0. Hence the maximal ideals
$$\mathfrak{m}_i' = \mathrm{Ker}(a_i' \otimes 1 : A \otimes_C k \to k)$$
of $A \otimes_C k$ are distinct from $\mathfrak{m} = \mathrm{Ker}(a_0 \otimes 1 : A \otimes_C k \to k)$. By Algebra, Lemma 00DS we would get an element $f \in A \otimes_C k$ with $f \in \mathfrak{m}$, but $f \notin \mathfrak{m}_i'$ for $i = 1, \ldots, n$. Consider the norm
$$g = \mathrm{Norm}_{s^\sharp \otimes 1}(t^\sharp \otimes 1(f)) \in A \otimes_C k$$
By Lemma 03BH this lies in the invariants $C^1 \subset A \otimes_C k$ of the base change groupoid (base change via the map $c : C \to k$). On the one hand, $a_1(g) \in k^*$ since the value of $t^\sharp(f)$ at all the points (which correspond to $b_1, \ldots, b_n$) lying over a_1 is invertible (insert future reference on property determinant here). On the other hand, since $f \in \mathfrak{m}$, we see that f is not a unit, hence $t^\sharp(f)$ is not a unit (as $t^\sharp \otimes 1$ is faithfully flat), hence its norm is not a unit (insert future reference on property determinant here). We conclude that C^1 contains an element which is not nilpotent and not a unit. We will now show that this leads to a contradiction. Namely, apply Lemma 03BK to the map $c : C \to C' = k$, then we see that the map of k into the invariants C^1 is injective and moreover, that for any element $x \in C^1$ there exists an integer $n > 0$ such that $x^n \in k$. Hence every element of C^1 is either a unit or nilpotent.

We still have to finish the proof of (1). We already know that $|U| \to |M|$ is surjective. It is clear that $|U| \to |M|$ is $|R|$-invariant. Finally, suppose $u_0, u_1 \in U$ maps to the same point $m \in M$. Then the induced field extensions $\kappa(u_0)/\kappa(m)$ and $\kappa(u_1)/\kappa(m)$ are algebraic (as A is integral over C as used above). Hence if k

is an algebraic closure of $\kappa(m)$ then we can find $\kappa(m)$-embeddings $\overline{u}_0 : \kappa(u_0) \to k$ and $\overline{u}_1 : \kappa(u_1) \to k$. These determine k-valued points $\overline{u}_0, \overline{u}_1 \in U(k)$ mapping to the same point of $M(k)$. By part (2) we see that there exists a point $\overline{r} \in R(k)$ with $s(\overline{r}) = \overline{u}_0$ and $t(\overline{r}) = \overline{u}_1$. The image $r \in R$ of $\overline{r}$ is a point with $s(r) = u_0$ and $t(r) = u_1$ as desired. □